INTERNET PROGRAMMING AND MOBILE PROGRAMMING

PROJECT 3: Design and Implementation of a Crowdsensed Drive Test
Mobile Application for Coverage Prediction

TASK 3: REQUIREMENT ANALYSIS

GROUP 10

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TASK 3: REQUIREMENT ANALYSIS

1. Introduction

Requirement analysis is one of the most important phases in software engineering because it ensures that the requirements collected from stakeholders are properly reviewed, validated, prioritized, and transformed into a structured Software Requirements Specification. In this project, requirement analysis is performed for the Crowdsensed Drive Test Mobile Application for Coverage Prediction. The analysis is based on data collected through the Google Form survey questionnaire distributed to smartphone users.

The survey collected information regarding signal strength, internet speed, dropped calls, network usage patterns, willingness to share location data, and user readiness to participate in crowdsensing activities. The collected dataset contains twenty-four survey responses and twenty-one variables related to mobile network experiences and user expectations. The requirement analysis process focuses on reviewing the collected requirements, identifying inconsistencies, prioritizing requirements, classifying them into functional and non-functional requirements, validating them with stakeholders, and developing a Software Requirements Specification.

2. Review And Analysis Of Gathered Requirements

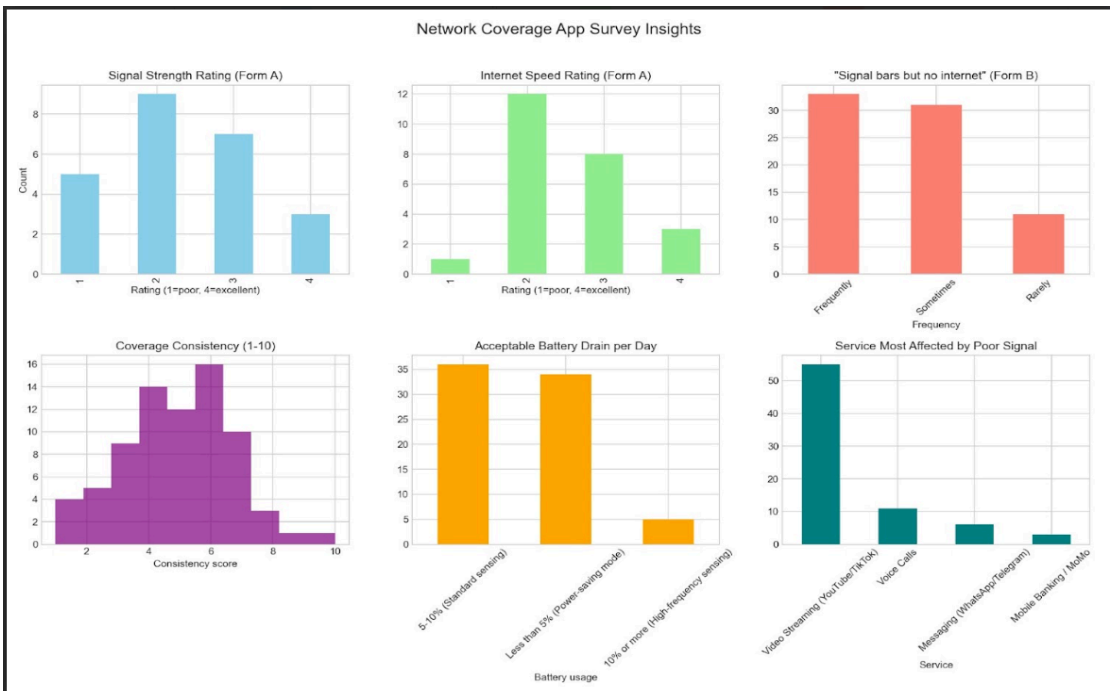
- The first stage of requirement analysis involves reviewing all requirements gathered through the survey form. The survey results indicate that most respondents use MTN and Orange networks, while some also use Camtel. Many respondents reported experiencing poor signal strength, dropped calls, and unstable internet speeds in rural areas, homes, and highly populated regions.

- Analysis of the survey data shows that several users rate their signal strength between one and three on a five-point scale, indicating weak or unstable connectivity. A significant number of respondents also report frequent signal loss and occasional dropped calls. This confirms the need for a system capable of monitoring network quality and identifying coverage holes.

- The gathered requirements were analyzed based on completeness, clarity, technical feasibility, and dependency relationships. The requirement for automatic collection of signal strength data was found to be technically feasible because smartphones provide APIs capable of accessing network measurements such as RSRP, RSSI, and SINR.

- Requirements related to offline storage and synchronization are also feasible because mobile applications can temporarily store data locally before transmitting it to the backend server when internet connectivity becomes available. The requirement for GPS integration depends on user permission and device capabilities.

- Dependency analysis shows that coverage prediction depends heavily on accurate data gathering, reliable GPS coordinates, and efficient backend analytics. Visualization features depend on proper data cleaning and mapping processes.



3. Identification Of Inconsistencies, Ambiguities And Missing Information

❖ Requirement analysis also involves identifying inconsistencies, ambiguities, and missing information within the collected dataset.

❖ Several inconsistencies were identified in the survey responses. For example, some respondents entered different spellings for similar values such as “Orange” and “Orannge.” Such inconsistencies affected data analysis and required standardization during data cleaning.

❖ Some responses contained missing location information, while others did not provide additional comments or detailed descriptions of poor coverage areas. These missing values reduce the completeness of the dataset and may limit accurate geographical analysis.

❖ Ambiguities were also observed in some responses. Certain respondents selected broad terms such as “Sometimes” or “Often” without specifying exact conditions or frequency ranges. This makes interpretation subjective and may require additional clarification through interviews or follow-up surveys.

❖ The analysis also identified missing technical information such as device specifications, operating system versions, and exact network technologies available in different regions. Including these details in future surveys would improve the quality of requirement analysis and support more accurate prediction models.

1. Prioritizing Requirements Based On Importance And Feasibility

✔ Requirement prioritization helps determine which system features are most important and should be implemented first based on stakeholder needs and technical feasibility.

- ✔ The highest priority requirement identified from the survey is automatic collection of network measurements such as **signal strength**, **network type, GPS coordinates, and timestamps**. This requirement forms the core functionality of the crowdsensed drive test system.

- ✔ Offline storage and synchronization were also identified as high-priority requirements because many respondents experience unstable connectivity. Without offline storage, valuable measurements could be lost when users move through weak coverage areas.

- ✔ Privacy protection and data anonymization are also high-priority requirements because survey results indicate that some users are concerned about sharing their location information. Implementing privacy controls is necessary to improve trust and encourage participation.

- ✔ Coverage map visualization and alerts for poor signal areas are medium-priority requirements because they improve user experience and provide useful insights for network operators.

- ✔ Additional features such as network speed tests and usage tracking are considered lower-priority requirements because they are supplementary features rather than core system functionalities.

4. Classification Of Requirements

Requirements identified during analysis are classified into functional and non-functional requirements.

Functional requirements:

- **Collecting Signal Measurement:** This requirement defines the ability of the mobile application to automatically gather network-related measurements from users' smartphones. The system should collect parameters such as RSSI, RSRP, RSRQ, SINR, network type, GPS coordinates, and timestamps. These measurements are essential because they help determine the quality of mobile network coverage in different geographical locations.

- **Storing Data Locally:** The system must support local data storage on the user's smartphone. This functionality is important because many users may experience unstable or unavailable internet connectivity while traveling through poor coverage areas. When the application cannot connect to the backend server, collected measurements should be temporarily stored inside a local database or storage system on the device. The system should ensure that no collected data is lost during offline operation.

- **Synchronizing Measurements With Backend Server:** The synchronization requirement ensures that measurements collected by the mobile application are transmitted to the centralized backend server. Synchronization occurs whenever internet connectivity becomes available. The synchronization process should upload locally stored measurements securely and efficiently. The backend server receives the data, validates it, cleans inconsistencies, and stores it inside a central

database for analysis.

- **Analyzing Network Coverage:** The system must analyze collected measurements in order to determine the quality of network coverage across different regions. This requirement involves processing signal strength data, identifying weak coverage areas, and detecting connectivity problems such as dropped calls or unstable internet performance.

- **Generating Report:** The system shall generate reports based on analyzed network measurements. Reports may include summaries of signal quality, connectivity statistics, dropped call frequencies, internet performance, and geographical coverage trends. Generated reports help administrators, researchers, and network operators understand the state of mobile network performance. Reports may be produced in textual, graphical, or tabular formats depending on user needs.

- **Visualizing Coverage Map:** The system should provide visual representations of collected network data through coverage maps. Visualization allows users and administrators to easily understand signal quality distribution across different locations. Coverage maps may use colors, symbols, or heat-map techniques to represent strong and weak network areas. For example, green regions may indicate strong coverage while red regions indicate poor signal quality.

Non-functional requirements:

✔ **Scalabilty:** Scalability refers to the ability of the system to handle increasing numbers of users, devices, and collected measurements without performance degradation. As the crowdsensing system grows, the backend infrastructure should be capable of processing large datasets generated by thousands of smartphones. The database and server architecture must therefore support expansion and efficient resource management.

Scalability is important because crowdsensed systems depend on participation from many users. A scalable system ensures long-term usability and supports future growth without requiring complete redesign.

✔ **Security:** Security ensures that collected data and system resources are protected against unauthorized access, attacks, and misuse. Since the application collects sensitive information such as location data, strong security measures are necessary. The system should use secure communication protocols such as HTTPS and encryption methods to protect transmitted data. Authentication and access control mechanisms should restrict backend access to authorized administrators only.

✔ **Reliability:** Reliability refers to the ability of the system to operate continuously and correctly without frequent failures. The application should consistently collect, store, synchronize, and process network measurements even under unstable network conditions. Reliable systems minimize data loss and ensure that collected measurements remain accurate and available. Error handling and recovery mechanisms should be implemented to manage synchronization failures

and unexpected application crashes.

- ✔ **Usability:** Usability describes how easy and user-friendly the application is for end users. The mobile application should provide a simple interface that allows users to interact with the system without requiring advanced technical knowledge. The interface should contain clear menus, readable text, and intuitive navigation. Users should easily understand how the application works, how data is collected, and how privacy settings are managed.

- ✔ **Battery Optimization:** Battery optimization ensures that the application minimizes smartphone power consumption during operation. Continuous sensing and GPS tracking may consume significant battery resources if not properly managed. The system should optimize sensing intervals, background operations, and synchronization frequency in order to reduce energy consumption. Efficient battery management is important because users may uninstall applications that excessively drain device batteries.

- ✔ **Efficient Data Synchronization:** Efficient data synchronization ensures that collected measurements are transferred quickly and accurately between smartphones and the backend server. The synchronization mechanism should minimize bandwidth usage, avoid duplicate data transmission, and reduce synchronization delays. Compression techniques and optimized communication protocols may be used to improve synchronization efficiency.

- ✔ **Privacy Protection:** Privacy protection ensures that user information and collected measurements are handled responsibly and ethically. Since the system collects location data, users may be concerned about how their information is used and stored. The system should anonymize collected data so that individual users cannot easily be identified. User consent should be obtained before collecting sensitive information such as GPS coordinates.

The survey data strongly supports the importance of non-functional requirements. Many respondents expressed concerns regarding privacy, battery consumption, and background data collection. This demonstrates that system usability and trust are equally important as technical functionality.

5. Software Requirement Specifications

5.1 .1 Introduction

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Crowdsensed Drive Test Mobile Application for Coverage Prediction. The system is designed to use smartphones as sensing devices to collect mobile network measurements such as signal strength, internet performance, GPS coordinates, and network quality indicators. The collected data is transmitted to a centralized backend server where it is analyzed to predict network coverage quality.

5.1.2 Purpose:

The purpose of this project is to design and implement a crowdsensed mobile application capable of collecting cellular network measurements from smartphone users and transmitting the collected data to a centralized backend system for coverage analysis and prediction.

5.1.3 Scope:

The system supports data collection, offline storage, synchronization, backend analytics, coverage prediction, report generation, and map visualization. The application is intended to help researchers, developers, and network operators identify weak coverage areas and optimize network performance.

5.1.4 Objectives

- To collect real-time mobile network measurements from smartphone users.
- To analyze network quality using crowdsensed data.
- To predict mobile coverage conditions in different locations.
- To visualize network performance through maps and reports.
- To provide a scalable and secure crowdsensing platform.

5.2 Overall Description

5.2.1 System Overview:

The Crowdsensed Drive Test Application consists of a mobile application, a backend server, a centralized database, and an analytics platform. Smartphone users act as sensing nodes that continuously collect network measurements. The collected information is synchronized with the backend server where it is analyzed and visualized.

5.2.2 Stakeholders

- **Smartphone Users:** Provide crowdsensed measurements through the mobile application.
- **Network Operators:** Use analyzed data to improve mobile network infrastructure.
- **Developers:** Design, maintain, and improve the software system.
- **Researchers:** Analyze collected datasets for academic and technical studies.
- **Administrators:** Manage backend systems, reports, and analytics dashboards.

5.2.3 Assumptions And Dependencies

The system assumes that users possess Android smartphones with GPS functionality and internet connectivity. The application depends on smartphone APIs for collecting network measurements and on stable backend infrastructure for analytics and data processing.

5.2 Funtional Requirements

- **Signal Measurement Collection:** The system shall collect signal strength indicators such as RSSI, RSRP, RSRQ, SINR, network type, GPS coordinates, and timestamps automatically from smartphones.

- **Local Data Storage:** The mobile application shall temporarily store collected measurements locally whenever internet connectivity is unavailable.

- **Data Synchronization:** The system shall synchronize locally stored measurements with the backend server once connectivity becomes available.

- **Coverage Analyses:** The backend system shall process collected measurements to identify weak coverage regions and analyze network quality.
- **Coverage Prediction:** The system shall use analytical models to predict mobile network coverage conditions across different geographical regions.

- **User Authentication:** The system shall support secure login and administrator authentication mechanisms.

- **Report Generation:** The system shall generate reports showing signal quality statistics, connectivity trends, and coverage summaries.

- **Coverage Visualization:** The system shall display analyzed network measurements using maps, charts, and graphical visualizations.

- **Notifications And alerts:**

The application shall notify users or administrators when synchronization fails or when poor signal areas are detected.

5.2.4 Non Functional Requirement:

- ✔ **Scalability:** The backend infrastructure shall support large-scale crowdsensed datasets generated by multiple users simultaneously.

- ✔ **Security:** The system shall use secure communication protocols and encryption mechanisms to protect collected data.

- ✔ **Reliability:** The system shall continue operating correctly even during unstable connectivity conditions.

- ✔ **Usability:** The mobile application shall provide a simple, intuitive, and user-friendly interface.

- ✔ **Battery Optimization:** The application shall minimize battery consumption during sensing and synchronization activities.

- ✔ **Performance:** Synchronization and analytics operations shall execute efficiently with minimal response delays.

✔ **Maintainability:** The software architecture shall support future updates and maintenance without affecting core functionality.

✔ **Privacy Protection:** The system shall anonymize sensitive user information and obtain user consent before collecting location data.

5.3.0 System Requiremnts:

5.3.1 Hardware Requirement:

- Android smartphone with GPS support
- Minimum 2GB RAM
- Internet connectivity (Wi-Fi or mobile data)
- Backend server infrastructure
- Cloud or local database server

5.3.2 Software Requirements

- Android operating system
- Mobile application development framework
- Backend server platform
- Database management system
- Analytics and visualization tools

5.3.3 Data Requirements: The system shall collect and manage technical network measurements including signal strength indicators, network provider information, timestamps, GPS coordinates, and internet performance data. All collected data shall be validated, cleaned, and standardized before analysis.

5.3.4 Security Requirements:

The system shall implement authentication and authorization mechanisms to restrict access to sensitive data and backend dashboards. Data transmitted between smartphones and backend servers shall be encrypted using secure communication protocols. User location data shall be anonymized to protect privacy.

5.3.5 User Interface Requirements:

The mobile application shall provide a clean and user-friendly interface with clear menus, navigation options, and visual indicators. The backend dashboard shall provide administrators with maps, charts, reports, and analytics summaries.

5.3.6 Constraints:

The accuracy of collected measurements may vary depending on smartphone hardware, GPS precision, environmental conditions, and network availability. Limited internet connectivity may also affect synchronization performance.

5.3.7 Testing Requirements

- Unit testing for mobile application functionalities
- Integration testing between mobile application and backend server
- Performance testing for synchronization and analytics
- Security testing for authentication and encryption mechanisms
- Usability testing with smartphone users

5.3.8 Future Enhancements:

Future improvements may include machine learning-based prediction models, support for iOS devices, advanced heat-map visualization, anomaly detection, and integration with big data analytics platforms.

5.3.9 Conclusions

This Software Requirements Specification document defines the requirements for the Crowdsensed Drive Test Mobile Application for Coverage Prediction. The document provides a strong foundation for system design, implementation, testing, deployment, and future maintenance. By combining crowdsensing techniques with mobile analytics, the system offers a scalable and cost-effective solution for mobile network monitoring and coverage prediction.

6. Validation Of Requirement With Stakeholders:

- Requirement validation ensures that the identified requirements accurately reflect stakeholder needs and expectations.

- Survey responses indicate that most users are willing to contribute anonymous network measurements if privacy and battery optimization concerns are addressed. This validates the feasibility of crowdsensing as a data collection approach.

- Validation is also performed through discussions with developers, researchers, and technical experts. These discussions confirm that smartphone APIs are capable of supporting the required measurements and that backend systems can process large-scale crowdsensed datasets.

- Stakeholder validation helps eliminate misunderstandings, confirm technical feasibility, and improve confidence in the proposed system design.

7. Conclusion:

- Requirement analysis plays a critical role in ensuring the success of the Crowdsensed Drive Test Mobile Application for Coverage Prediction. By analyzing survey responses and identifying stakeholder needs, developers gain a clear understanding of system objectives, technical requirements, and user expectations.

- The survey dataset provides valuable insights regarding signal quality, connectivity challenges, privacy concerns, and desired application features. Requirement analysis also helps identify inconsistencies, prioritize system features, classify requirements, validate stakeholder expectations, and support the development of a structured Software Requirements Specification.

- Overall, the requirement analysis process establishes a strong foundation for successful system implementation, testing, deployment, and future improvement.